25 There are some red counters and some white counters in a bag.
At the start, 7 of the counters are red, the rest of the counters are white.

Alfie takes at random a counter from the bag.
He does not put the counter back in the bag.
Alfie then takes at random another counter from the bag.

The probability that the first counter Alfie takes is white **and** the second counter Alfie takes is red is $\frac{21}{80}$

Work out the number of white counters in the bag at the start.

Let x be the whole number of white counters.

P(first white) = x/(x+7).

1 mark
Start with x whites

Then 7 reds remain among x+6 counters.

$$\frac{x}{x+7} \times \frac{7}{x+6} = \frac{21}{80}$$

1 mark
Without replacement multiply probabilities

$560x = 21(x+7)(x+6)$

$21x^2 - 287x + 882 = 0$; divide by 7:

$3x^2 - 41x + 126 = 0$

1 mark
Correct quadratic

×	$3x$	-14
x	$3x^2$	$-14x$
-9	$-27x$	126

$(3x - 14)(x - 9) = 0$

1 mark
Solve quadratic

$x=14/3$ or $x=9$.

A counter count is a whole number, so x = 9 whites.

1 mark
Reject 14/3; 9 whites

(Total for Question 25 is 5 marks)